

Building with AI – Mini Project

Project Scenario

You are a junior software developer working on a small internal tool for a startup.

The company needs a simple, functional web application to solve a real everyday problem such as managing tasks, notes, or basic user interactions. Your goal is to design, build, refine, and deploy a small application using AI as a development assistant.

Project Description

In this mini project, you will build a small but complete web application using HTML, CSS, JavaScript, and optionally a simple backend. AI tools will be used throughout the project to assist with planning, design, coding, refactoring, testing, and deployment.

The focus is on understanding the full development lifecycle and use the AI chatbot effectively to help you complete the project, **not on building a complex system.**

Project Options

You may choose ONE of the following project types (or a similar scope-approved idea):

- To-Do App
- Notes App
- Task Tracker
- Simple Shopping Cart
- Weather Dashboard
- Login/Register System

The chosen project must remain small, realistic, and beginner-friendly.

Project Requirements

Phase 1: Planning & Analysis

- Define the problem your app solves.
- Use AI to help select and validate the project idea.
- Identify core features only (avoid over-scoping).
- Define user stories using the format: As a [user], I want to [action], so I can [goal].
- Identify the data model (objects, fields, validations).
- Plan API endpoints if a backend is used.

Phase 2: Design

- Design a simple UI layout (wireframe or sketch).
- Plan folder structure using modern JavaScript conventions.
- Define components, services, and utilities.
- Use AI to validate structure and simplicity.

Phase 3: Development

- Create the project structure and initial files.
- Build the UI skeleton using HTML/CSS.
- Implement core functionality with JavaScript.
- Add event handling and basic validation.
- Optionally implement a simple Node.js backend.
- Use AI for code generation, comments, and explanations.

Phase 4: Refinement & Testing

- Ask AI to review and suggest improvements.
- Refactor code for readability and maintainability.
- Improve UI using simple CSS enhancements.
- Test edge cases, empty inputs, and user interactions.
- Create a testing checklist.

Phase 5: Deployment

- Prepare the project for production (clean code, organize files).
- Deploy the app using GitHub Pages or Netlify.
- Verify the deployed application works correctly.
- Provide a live deployment link.

Deliverables

- Source code repository (GitHub).
- Deployed application link.
- Short project README explaining the app, features, and usage.
- Screenshots of the application.

Project Submission

This project is for **practice only**. You are not **required to submit it**. You are encouraged to ask questions and seek guidance during your weekly live sessions with your instructor.

Self- Evaluation Rubric

Criteria	Excellent (4)	Good (3)	Needs Improvement (2)
Planning & Scope	Clear, well-scoped, excellent use of AI	Mostly clear, minor issues	Unclear or over-scoped
Design & Structure	Clean structure, logical design	Acceptable structure	Poor organization
Functionality	All features work correctly	Minor bugs	Major functionality issues
Code Quality	Readable, well-commented, clean	Some clarity issues	Hard to read, messy
Refinement & Testing	Well-refined and tested	Some testing done	Little to no testing
Deployment	Successfully deployed and accessible	Deployed with minor issues	Not deployed